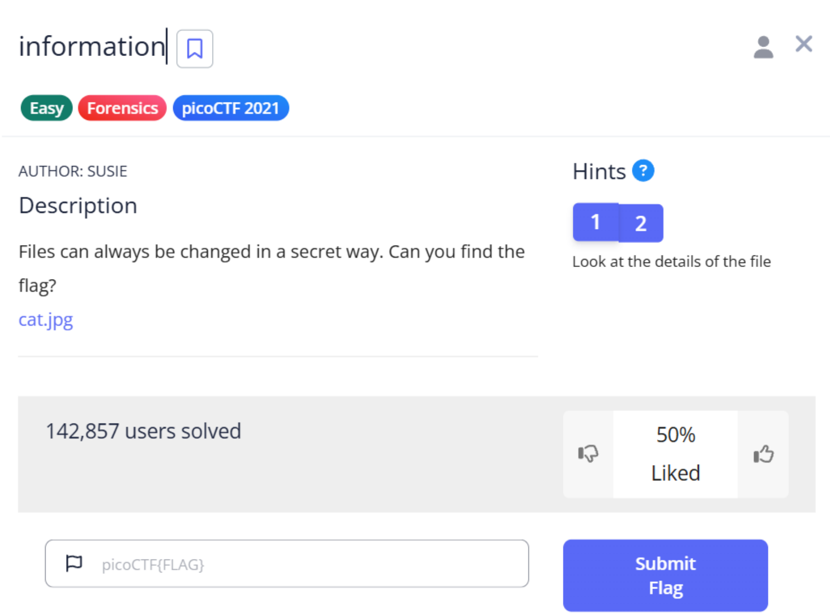
HERNANDO, PANTINE B. **BSIT-3**

PICO CTF-CHALLENGE 4



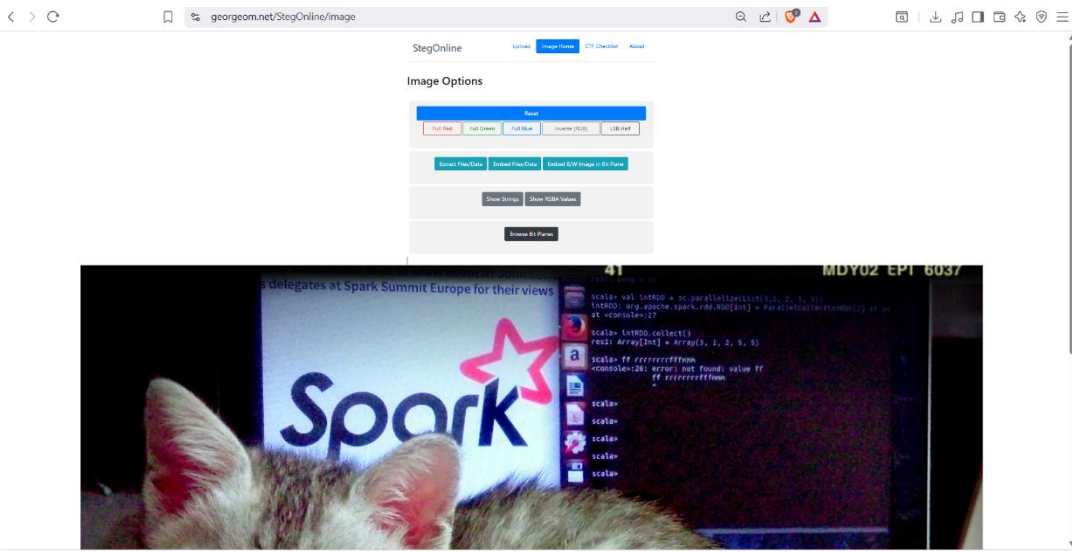
INFORMATION

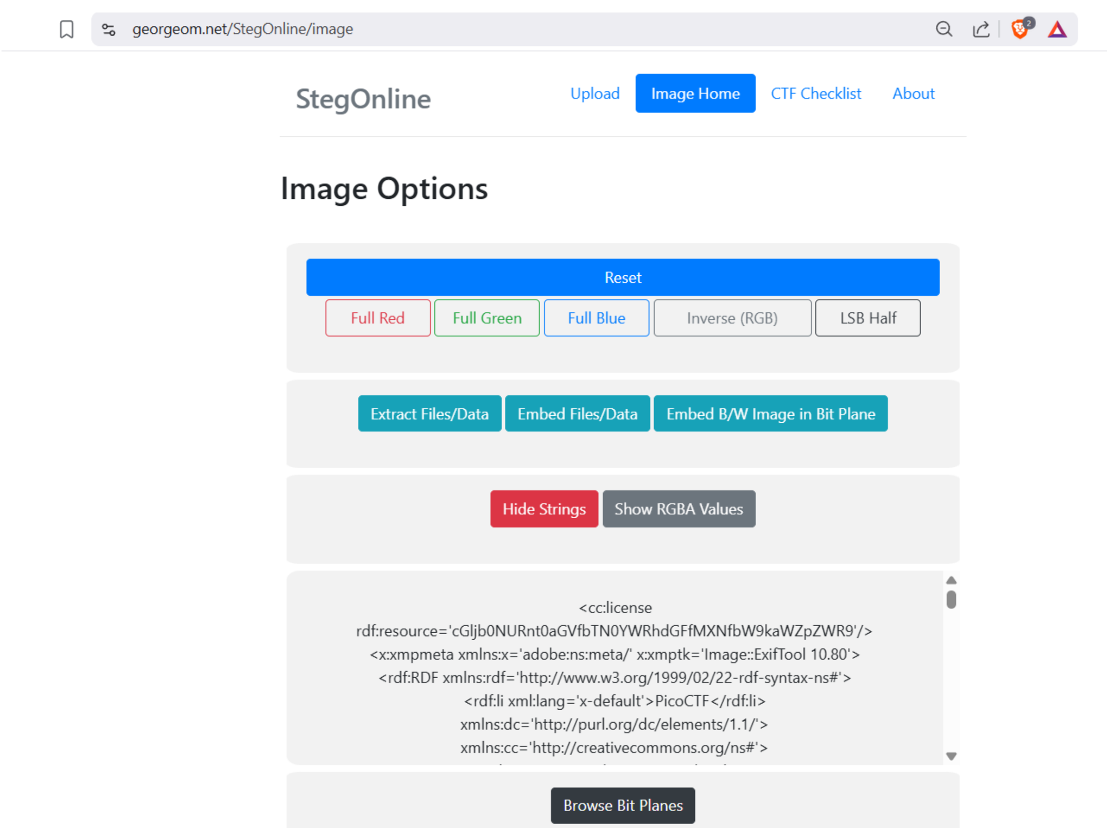
- This forensic challenge involves extracting hidden data from an image file named cat.jpg. The solution requires looking past the visual content to find an encoded string hidden in the file's internal properties.

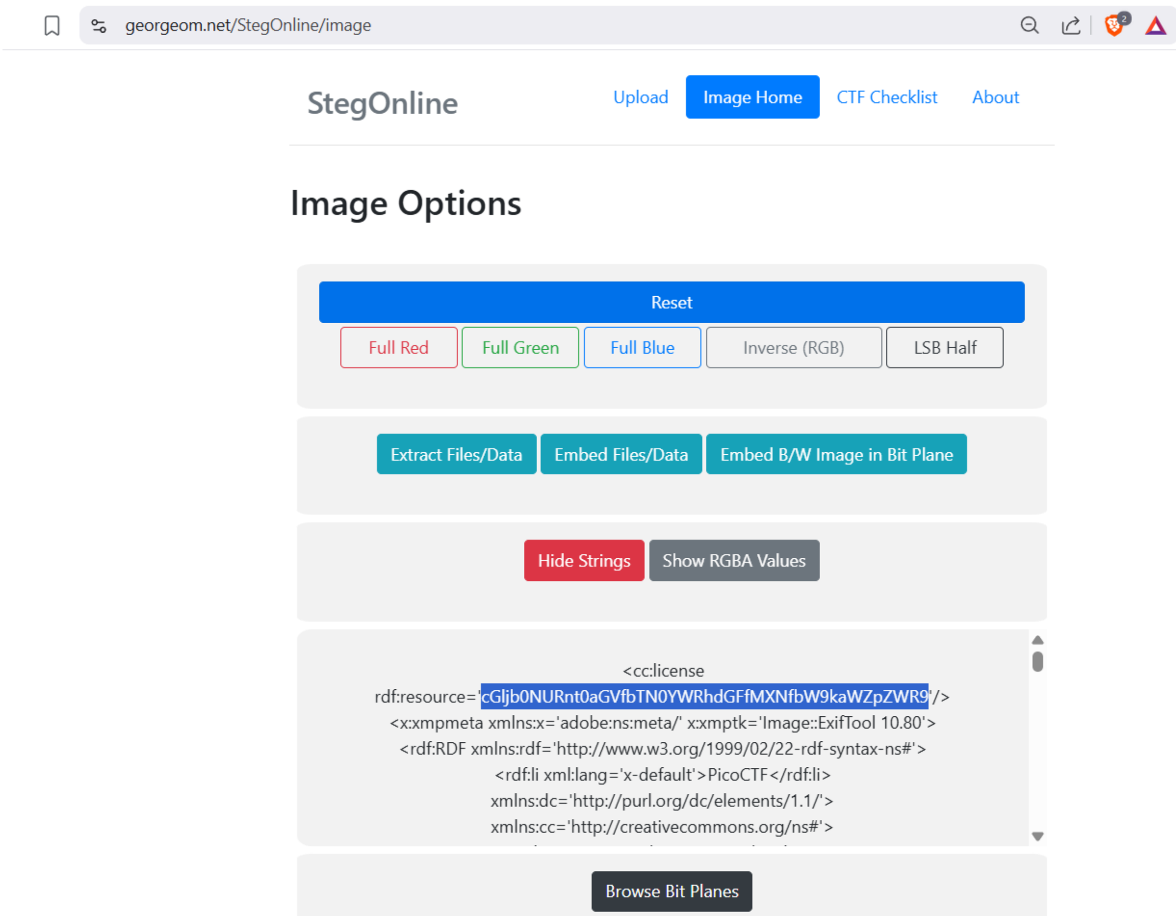
Steps:

- **Step 1: Analyze the Challenge File**: The investigator is provided with an image file of a kitten (cat.jpg) and a hint to "look at the details of the file".

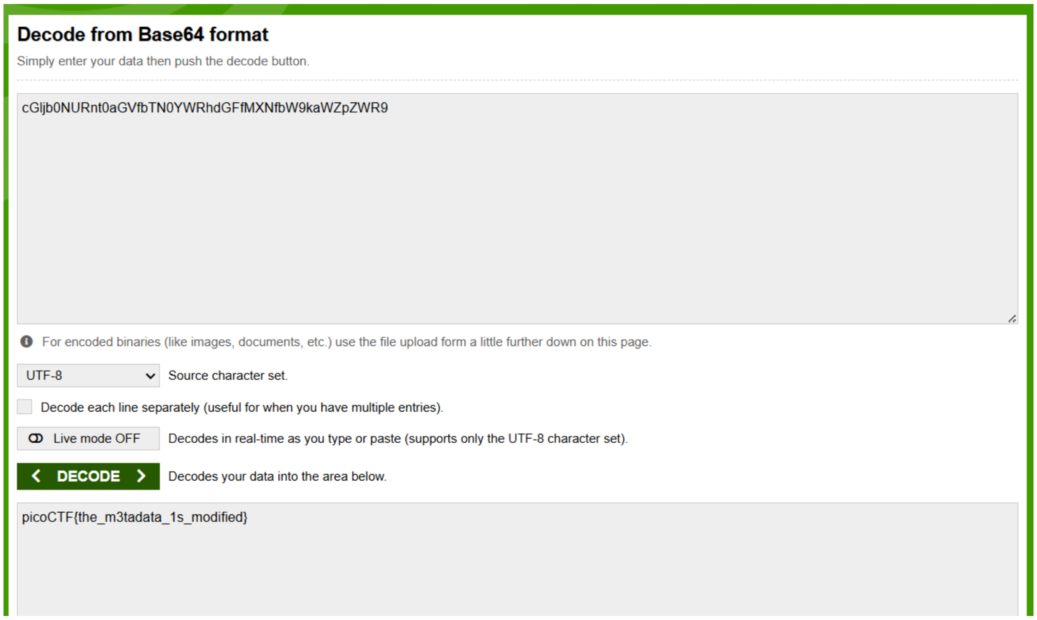
- **Step 2: Inspect File Metadata**: Using a steganography or metadata tool like **StegOnline**, the investigator examines the file's hidden attributes.







Step 3: Locate the Hidden String: By viewing the image options and strings, a suspicious license resource string is found: cGljb0NURnt0aGVfbTNOYWRhdGFfMXNfbW9kaWZpZWRSfQ==. Note: The provided source text displays the string as cGljb0NURnt0aGVfbTNOYWRhdGFfMXNfbW9kaWZpZWRS.



- **Step 4: Identify the Encoding**: The string consists of alphanumeric characters and appears to be **Base64** encoded, a common method for concealing data in CTF challenges.
- **Step 5: Decode the Payload**: The investigator enters the string into a **Base64 Decoder**.
- **Step 6: Retrieve the Final Flag**: Upon clicking decode, the plain text flag is revealed.
- **Final Flag:** picoCTF{th3_m3tadata_1s_modifi3d}

Reflection

This challenge reinforces the importance of **Metadata Forensics**. It demonstrates that digital files contain layers of information—such as the cc:license field—that are not immediately visible to the casual viewer. By modifying these standard fields, a user can hide sensitive information in plain sight.

For a forensic investigator, this highlights that every part of a file's header is a potential storage location for hidden data. Success in these challenges often depends on using the right tool (like ExifTool or StegOnline) to expose what standard image viewers ignore.